Syllabus Details

Syllabus ID:	11081
Syllabus Name:	**Data Mining_Khai phá dữ liệu**
Course Name English:	**Data Mining**
Subject Code:	**DBM301**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	
Description:	Data mining is a multidisciplinary field, drawing work from areas including database technology, machine learning, statistics, pattern recognition, information retrieval, neural networks, knowledge-based systems, artificial intelligence, high-performance computing, and data visualization. This course presents techniques for the discovery of patterns hidden in large data sets, focusing on issues relating to their feasibility, usefulness, effectiveness, and scalability. As a result, the course is a comprehensive introduction to data mining, presented with effectiveness and scalability issues in focus. It should be useful for computing science students, application developers, and business professionals at master degree, as well as researchers involved in any of the disciplines listed above.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Internet - Matlab trial version, Python
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	358/QĐ-ĐHFPT dated 04/03/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/3/2024

1 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Data Mining: Concepts and Techniques, Third Edition (The Morgan Kaufmann Series in Data Management Systems) http://library.books24×7.com	Jiawei Han, Micheline Kamber and Jian Pei	Morgan Kaufmann Publishers	2012	3rd	978-0-12-381479-1	True	False	True	

11 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe main concepts of Data mining
2	CLO2	Describe the Data
3	CLO3	Describe the Data Preprocessing
4	CLO4	Understand the Weka tool
5	CLO5	Describe the Data Warehouse & OLAP Technology
6	CLO6	Describe the Data Cube Technology
7	CLO7	Describe the Mining Frequent Patterns, Association Rule Mining and Correlation
8	CLO8	Apply the Classification
9	CLO9	Apply the Clustering
10	CLO10	Describe the Outlier Detection
11	CLO11	Understand Data Mining Trends and Research Frontiers

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction Introduction to Data mining	Offline	LO0		Chapter 1 Slide 1	DBM301	Read: Chapter 1 Slide 1	
2	Course Introduction Introduction to Data mining	Offline	LO0		Chapter 1 Slide 1	DBM301	Read: Chapter 1 Slide 1	
3	Getting to Know Your Data	Offline	LO1		Chapter 2 Slide 2		Read: Chapter 2 Slide 2	
4	Getting to Know Your Data	Offline	LO1		Chapter 2 Slide 2		Read: Chapter 2 Slide 2	
5	Data Preprocessing	Offline	LO2		Chapter 3 Slide 3		Read: Chapter 3 Slide 3	
6	Data Preprocessing	Offline	LO2		Chapter 3 Slide 3		Read: Chapter 3 Slide 3	
7	Data Preprocessing	Offline	LO2		Chapter 3 Slide 3		Read: Chapter 3 Slide 3	
8	Data Preprocessing	Offline	LO2		Chapter 3 Slide 3		Read: Chapter 3 Slide 3	
9	Introduction to Weka & Data Prepocessing	Offline	LO3		Web of Weka		Web of Weka	
10	Introduction to Weka & Data Prepocessing	Offline	LO3		Web of Weka		Web of Weka	
11	Introduction to Weka & Data Prepocessing	Offline	LO4		Web of Weka			
12	Introduction to Weka & Data Prepocessing	Offline	LO4		Web of Weka			
13	Data Warehouse & OLAP Technology	Offline	LO4		Chapter 4 Slide 4		Read: Chapter 4 Slide 4	
14	Data Warehouse & OLAP Technology	Offline	LO4		Chapter 4 Slide 4		Read: Chapter 4 Slide 4	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
15	Data Warehouse & OLAP Technology	Offline	LO4		Chapter 4 Slide 4		Read: Chapter 4 Slide 4	
16	Data Warehouse & OLAP Technology	Offline	LO4		Chapter 4 Slide 4		Read: Chapter 4 Slide 4	
17	Check Assignment 1	Offline					Presentation	
18	Check Assignment 1	Offline					Presentation	
19	Check Assignment 1	Offline					Presentation	
20	Check Assignment 1	Offline					Presentation	
21	Data Cube Technology	Offline	LO5		Chapter 5 Slide 5		Read: Chapter 5 Slide 5	
22	Data Cube Technology	Offline	LO5		Chapter 5 Slide 5		Read: Chapter 5 Slide 5	
23	Data Cube Technology	Offline	LO5		Chapter 5 Slide 5		Read: Chapter 5 Slide 5	
24	Data Cube Technology	Offline	LO5		Chapter 5 Slide 5		Read: Chapter 5 Slide 5	
25	Mining Frequent Patterns, Association Rule Mining and Correlation	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
26	Mining Frequent Patterns, Association Rule Mining and Correlation	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
27	Mining Frequent Patterns, Association Rule Mining and Correlation	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
28	Mining Frequent Patterns, Association Rule Mining and Correlation	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
29	Exercise 1	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
30	Exercise 1	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
31	Exercise 1	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
32	Exercise 1	Offline	LO6		Chapter 6,7 Slide 6,7		Read: Chapter 6,7 Slide 6,7	
33	Classification	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
34	Classification	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
35	Classification	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
36	Classification	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
37	Exercise 2	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
38	Exercise 2	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
39	Exercise 2	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
40	Exercise 2	Offline	LO7		Chapter 8,9 Slide 8,9		Read: Chapter 8,9 Slide 8,9	
41	Cluster	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
42	Cluster	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
43	Cluster	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
44	Cluster	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
45	Exercise 3	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
46	Exercise 3	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
47	Exercise 3	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
48	Exercise 3	Offline	LO8		Chapter 10,11 Slide 10,11		Read: 'Chapter 10,11 Slide 10,11	
49	Outlier Detection	Offline	LO9		Chapter 12 Slide 12		Read: Chapter 12 Slide 12	
50	Outlier Detection	Offline	LO9		Chapter 12 Slide 12		Read: Chapter 12 Slide 12	
51	Data Mining Trends and Research Frontiers	Offline	LO10		Chapter 13 Slide 13		Read: Chapter 13 Slide 13	
52	Data Mining Trends and Research Frontiers	Offline	LO10		Chapter 13 Slide 13		Read: Chapter 13 Slide 13	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
53	Exercise 4	Offline	LO10		Chapter 13 Slide 13		Read: Chapter 13 Slide 13	
54	Exercise 4	Offline	LO10		Chapter 13 Slide 13		Read: Chapter 13 Slide 13	
55	Exercise 4	Offline	LO10		Chapter 13 Slide 13		Read: Chapter 13 Slide 13	
56	Exercise 4	Offline	LO10		Chapter 13 Slide 13		Read: Chapter 13 Slide 13	
57	Check Assignment 2	Offline					Presentation	
58	Check Assignment 2	Offline					Presentation	
59	Check Assignment 2	Offline					Presentation	
60	Check Assignment 2	Offline					Presentation	

8 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1	Why Data Mining?
2	1	CQ2	Why Data Mining?
3	1	CQ3	What Kinds of Data Can Be Mined?
4	2	CQ1	What Kinds of Patterns Can Be Mined?
5	2	CQ2	Which Technologies Are Used?
6	2	CQ3	Which Kinds of Applications Are Targeted?
7	3	CQ1	What is Data Objects?
8	3	CQ2	What is Attribute Types?

3 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	30.0%	>0	90' in class		Theory present or project	N/A	technical aspect of studied topics	- Guided by instructor - Application or Theory - Work in group	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	Exercise	on-going	4	40.0%	>0	10'-30'		Any appropriate type	10		in class, by instructor	Instruction and schedule for Progress tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has resposibility to review the test for students after graded.
3	Final exam	Final exam	1	30.0%	4	30'		Multiple choicesMarked by Computer	30	concepts, algorithms; all studied chapters	by exam board, using computer	The exam questions must be updated or different at least 70% to the previous ones.